Project 3 Report — A/B Testing with Shiny and Google Analytics

STAT5243 Project 3

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Deployed Shiny App: Deployed Shiny App: https://baixuanchen5243.shinyapps.io/5243-project3/

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1 Repository Structure

The GitHub repository contains the Shiny application source code, A/B testing data, analysis notebook, and exported summary files. The main files are summarized below.

Table 1: Repository Structure and File Descriptions

File	Description
`app.R`	Main Shiny application file. It contains the app interface, server logic, A/B assignment, Google Analytics 4 event tracking, and user interaction logging.
`ab_metrics.csv`	Main exported A/B testing dataset. It records user interaction events such as demo CTA impressions, button clicks, dataset loading, first action completion, and downstream engagement events.
`ab_session_summary.csv`	Session-level summary file created from the event-level data. It aggregates events by session and variant so that each session is counted at most once for each outcome.
`ab_event_counts_by_variant.csv`	Event count summary by A/B variant. This file is used to describe how often each tracked event occurred under Version A and Version B.
`ab_test_summary_by_variant.csv`	Summary table of A/B testing metrics by variant, including session counts, click-through rate, demo dataset load rate, first action completion rate, and downstream engagement rate.
`ab_test_analysis.ipynb`	Python analysis notebook used to clean and summarize the exported GA4 data, construct session-level metrics, run Fisher's exact tests, and generate the final statistical results.
`datadog_consolidated_wide.csv`	Supporting monitoring or interaction log file exported from the app instrumentation system. It provides additional behavioral tracking information but is not the primary dataset for the A/B statistical test.
`README.md`	Project overview file. It summarizes the purpose of the Shiny app, how to run the project, the experiment design, and the structure of the repository.

Overall, the repository is organized to make the project reproducible. The file `app.R` defines the experiment and collects behavioral data, while the CSV files and analysis notebook document how the raw event data were converted into session-level metrics for statistical testing.

2 Introduction and Research Question

Experiments and A/B tests are widely used to evaluate whether a design intervention causes a measurable behavioral change. In this project, we implemented an A/B test in a Shiny-based Data Explorer application. The purpose of the experiment was to assess whether changing the wording of a demo dataset call-to-action (CTA) button would affect user engagement and downstream task completion.

The demo dataset CTA is one of the most important onboarding elements in the app because it provides the fastest way for a new user to start interacting with the system. Since early-stage interaction is critical in web applications, we focused on testing whether a small wording change

could improve engagement. Specifically, the experiment compares the original button label, "Load Demo Dataset," with the revised label, "Try Demo Dataset Instantly."

Our research question was:

> Does changing the demo CTA text from **"Load Demo Dataset"** to **"Try Demo Dataset Instantly"** increase the session-level demo dataset load rate in the Shiny app?

The hypotheses were defined as follows:

- **Null hypothesis (H_0):** The button wording has no effect on the session-level demo dataset load rate. In other words, the demo dataset load rate is the same for Version A and Version B.
- **Alternative hypothesis (H_1):** The treatment wording, **"Try Demo Dataset Instantly,"** is associated with a higher session-level demo dataset load rate than the control wording, **"Load Demo Dataset."**

This question is practically meaningful because the CTA is directly tied to the first user action in the application. If a wording change increases demo dataset usage, it may also increase the likelihood that users continue to later stages of the app, such as data preview, cleaning, feature engineering, and exploratory analysis.

Google Analytics 4 was used to track user behavior, including page views, button clicks, and other interaction events. The collected data was exported as a CSV file and analyzed using Python. The main objective of this report is to compare user engagement between the two versions and determine whether the button text change produced a meaningful difference in user behavior.

3 Experimental Design and Methodology

3.1 A/B Testing Setup

A/B testing is a controlled experiment used to compare two versions of a product or interface. In this project, Version A serves as the control group, while Version B serves as the treatment group.

The purpose of the experiment is to test whether the modified version of the app improves user engagement compared with the original version.

The general design is:

1. Users interact with the Shiny app.
2. Each session is assigned to either Version A or Version B.
3. Google Analytics records user events.
4. The exported data is analyzed to compare engagement metrics across groups.

3.2 Control and Treatment Groups

The experiment compares two versions of the Shiny application that differ in the wording of the demo dataset button. The goal is to evaluate whether a more action-oriented button label increases user engagement with the demo dataset feature.

- **Control group: Version A — "Load Demo Dataset"**
  Version A is the baseline version of the application. In this version, the button is labeled "Load Demo Dataset." This wording is direct and functional, but it is relatively neutral. Users assigned to Version A saw this original button label when interacting with the app. Their behavior was used as the reference group for evaluating the effect of the wording change.

- **Treatment group: Version B — "Try Demo Dataset Instantly"**
  Version B is the treatment version of the application. In this version, the button label was changed to "Try Demo Dataset Instantly." This wording is more action-oriented and emphasizes immediacy. The phrase "Try" may make the action feel easier and less formal, while "Instantly" suggests that the user can quickly explore the demo dataset without extra setup. Users assigned to Version B saw this modified button label.

The main purpose of this A/B test is to determine whether the revised button wording in Version B leads to higher engagement than the original wording in Version A. Specifically, we compare metrics such as the number of button clicks, click-through rate, average clicks per session, and other session-level engagement measures between the two groups. If Version B produces higher engagement, it may suggest that clearer and more action-oriented wording improves user interaction with the app.

By comparing these two groups, we can evaluate whether the design change affects user behavior.

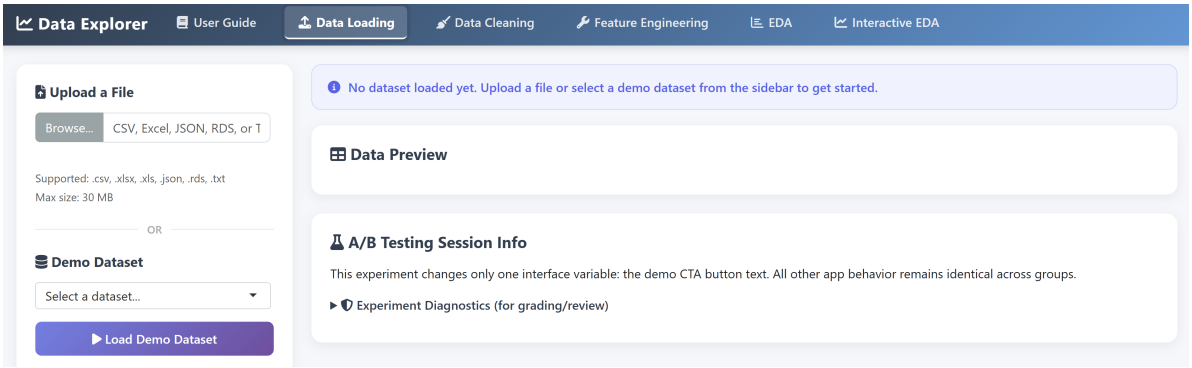

Figure 1: Control Version A: "Load Demo Dataset".

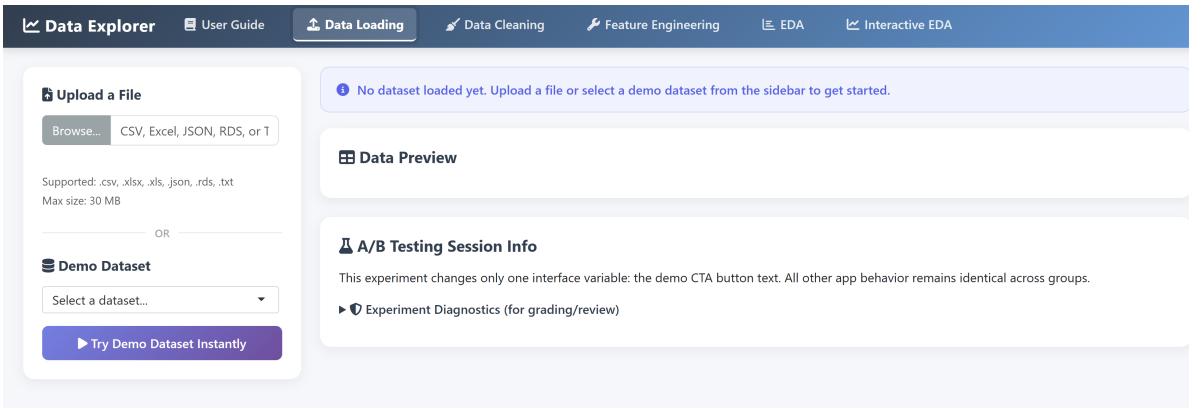

Figure 2: Treatment Version B: "Try Demo Dataset Instantly".

3.3 Outcome Metrics

Because the treatment changes the wording of the demo dataset call-to-action button, the main outcome metric should directly measure whether users are more likely to interact with that button. We therefore separate the metrics into one primary metric and several secondary engagement metrics.

Primary Metric

- **Demo dataset load rate**
  This is the main metric for the experiment. It measures the proportion of CTA-exposed sessions in which users successfully loaded a demo dataset. This metric was selected because it directly captures whether the demo dataset CTA encouraged users to begin using the app. In this app logic, the session-level click-through rate and demo dataset load rate were identical because sessions with a demo button click also triggered a successful demo dataset load.

Secondary Metrics

The secondary metrics are used to evaluate broader user engagement after the initial button exposure:

- **Number of sessions**
  This measures how many user sessions were assigned to each version. It is mainly used to check whether the A/B split is reasonably balanced.
- **Total number of events**
  This measures the total amount of tracked activity in each version. Since total events can be affected by the number of sessions, it should be interpreted together with session counts.
- **Average events per session**
  This measures the average level of interaction within each session. It is more comparable than total event count because it adjusts for differences in group size.
- **Average clicks per session**
  This measures how actively users clicked within the app after being assigned to a version.
- **Downstream user actions**
  This measures whether users continued to later stages of the app after the demo dataset interaction, such as data preview, data cleaning, feature engineering, or exploratory analysis.
- **Target event rate**
  This measures the proportion of sessions that completed a predefined target action, such as clicking the demo dataset button or reaching a later app module.

Together, these metrics allow us to evaluate both the direct effect of the button wording change and its broader effect on user engagement. The primary metric focuses on the immediate CTA response, while the secondary metrics describe whether users continue interacting with the application after the initial action.

4 Data Collection

4.1 Google Analytics 4 Integration

Google Analytics 4 (GA4) was integrated into the Shiny app to collect user interaction data from real app sessions. The app sends event information to GA4 when users visit the page, switch tabs, click the demo dataset button, load a dataset, or interact with later-stage app features.

The main tracked events used in the analysis were:

- `session_started`: records the beginning of a user session.
- `tab_viewed`: records when a user views a different app tab.
- `demo_cta_impression`: records when the demo dataset call-to-action is shown.
- `demo_button_clicked`: records when the user clicks the demo dataset button.
- `demo_dataset_loaded`: records when a demo dataset is successfully loaded.
- `first_action_completed`: records when the user completes the first meaningful action in the app.
- `cleaning_options_applied`: records whether the user applied data cleaning options.
- `feature_engineering_applied`: records whether the user used a feature engineering operation.
- `eda_plot_generated`: records whether the user generated an interactive EDA plot.

Each event also included experiment-related parameters such as the assigned variant, button text, session identifier, dataset name, and tab name. These parameters allowed us to compare user behavior between Version A and Version B at the session level.

4.2 Data Collection and Exported Dataset

The dataset used in this report was exported from Google Analytics 4 and processed into CSV files for analysis. The main file used for session-level analysis was `ab_metrics.csv`. Each row represents a tracked user event from the Shiny app. The important variables used in the analysis are described in Table 2.

Table 2: Important Variables in the A/B Testing Dataset

Variable	Description
`session_id`	Identifier for a user session. This allows events from the same session to be grouped together.
`variant`	A/B test group assignment, either Version A or Version B.
`condition`	Descriptive label for the experimental condition, such as Control A or Treatment B.
`button_text`	Text shown on the demo dataset button. Version A used "Load Demo Dataset," while Version B used "Try Demo Dataset Instantly."
`event` or `event_name`	Name of the tracked user event, such as `demo_button_clicked` or `demo_dataset_loaded`.
`event_count`	Number of times a specific event occurred.
`timestamp`	Time when the event was recorded.

For statistical testing, the event-level data were converted into session-level indicators, so each session was counted at most once for each outcome event.

5 Exploratory Data Analysis

Exploratory data analysis was used to understand the structure of the collected A/B testing data before formal hypothesis testing. The raw data were exported from Google Analytics 4 and then converted into session-level summaries. Each row in the raw dataset represents a tracked user event, including the session identifier, event name, assigned variant, button text, dataset name, tab name, elapsed time, and timestamp.

5.1 Visual Summary of A/B Testing Data

To better understand the collected A/B testing data, we created several exploratory visualizations. These figures summarize raw event counts, funnel behavior, metric rates, the primary outcome, time to first action, and session-level outcome rates.

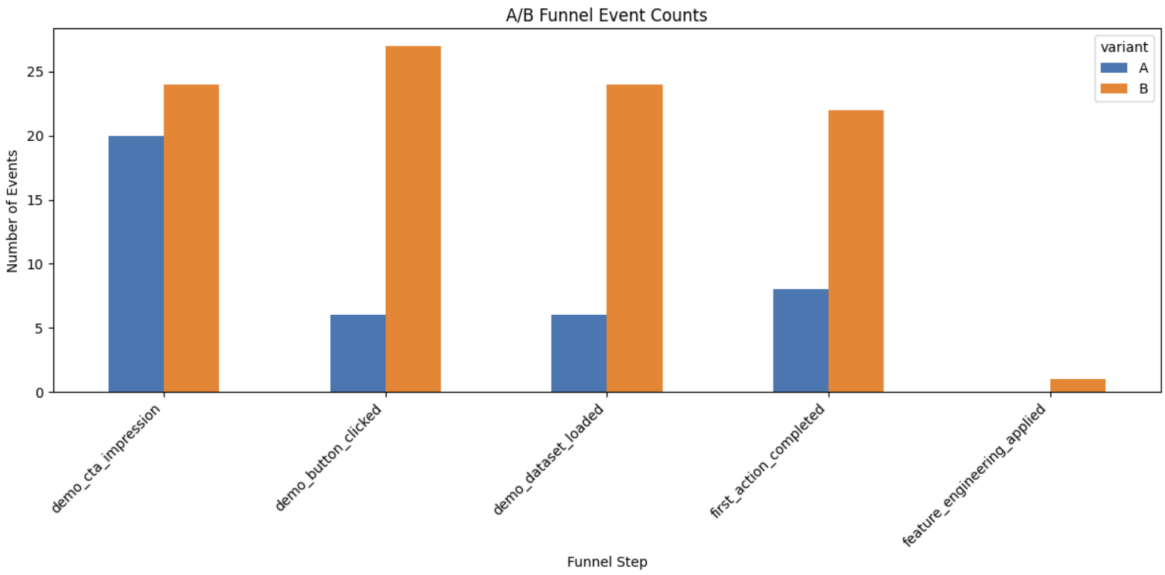

Figure 3: Event Counts by A/B Variant

Figure 3 compares the raw number of tracked events between Version A and Version B. Version B generated more interaction events for most categories, especially `demo_button_clicked`, `demo_dataset_loaded`, `eda_plot_generated`, and `tab_viewed`.

Interpretation: This suggests that users in Version B interacted more frequently with the app overall. However, because these are raw event counts, they may include repeated actions from the same session and should be interpreted descriptively rather than as formal conversion rates.

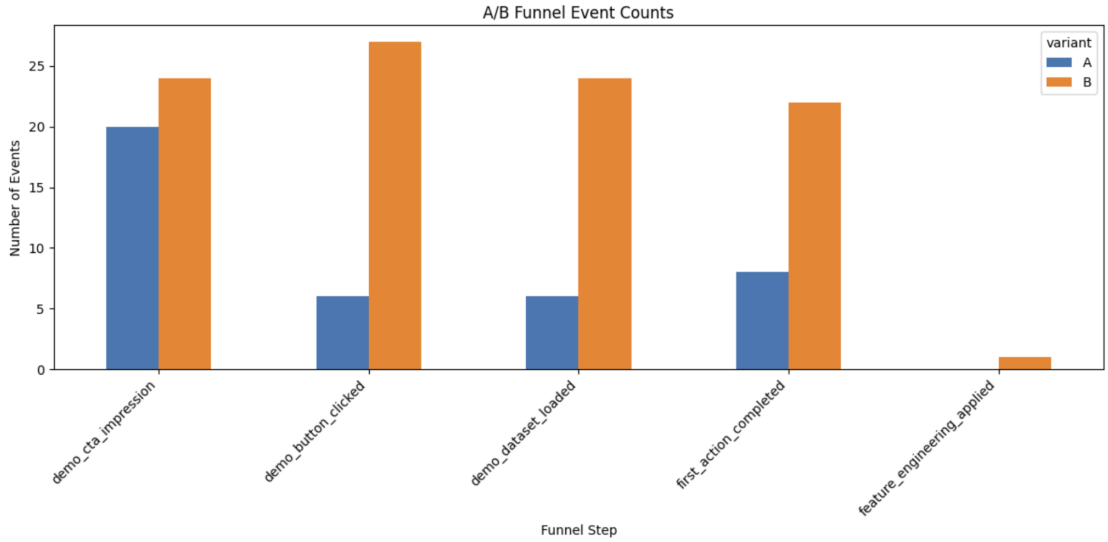

Figure 4: A/B Funnel Event Counts

Figure 4 focuses on key funnel events in the user journey. Both groups had similar numbers of demo CTA impressions, but Version B had much higher counts for demo button clicks, demo dataset loads, and first action completions.

Interpretation: The funnel pattern suggests that the treatment button text helped move users from simply seeing the CTA to actively clicking and loading a demo dataset. This supports the idea that Version B improved early-stage engagement.

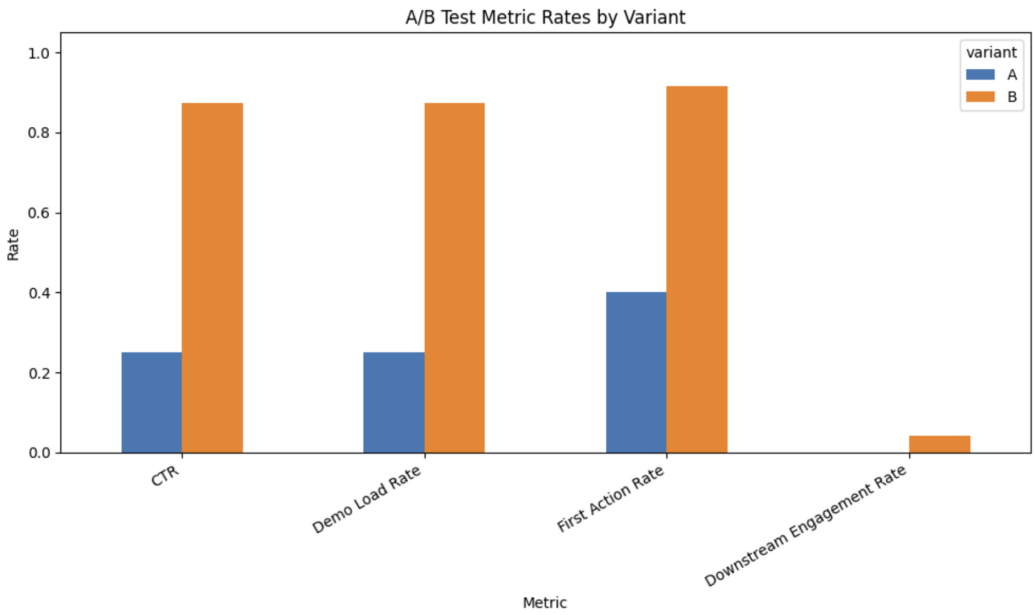

Figure 5: A/B Test Metric Rates by Variant

Figure 5 compares session-level engagement rates between the two variants. Version B had higher click-through rate, demo load rate, and first action completion rate than Version A.

Interpretation: The largest improvements appear in CTR and demo load rate, which are directly related to the changed button text. This provides descriptive evidence that the phrase "Try Demo Dataset Instantly" was more effective than "Load Demo Dataset" in encouraging users to start using the app.

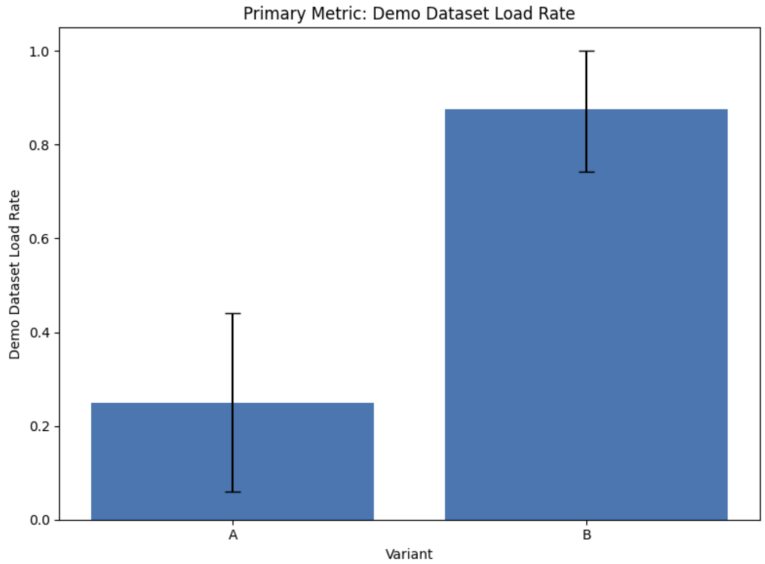

Figure 6: Primary Metric: Demo Dataset Load Rate

Figure 6 shows the primary metric, demo dataset load rate, for Version A and Version B. Version B had a substantially higher demo load rate than Version A.

Interpretation: This figure visually supports the main statistical result: Version B strongly outperformed Version A on the primary metric. The difference is large enough to suggest a meaningful practical improvement, not just a small numerical change.

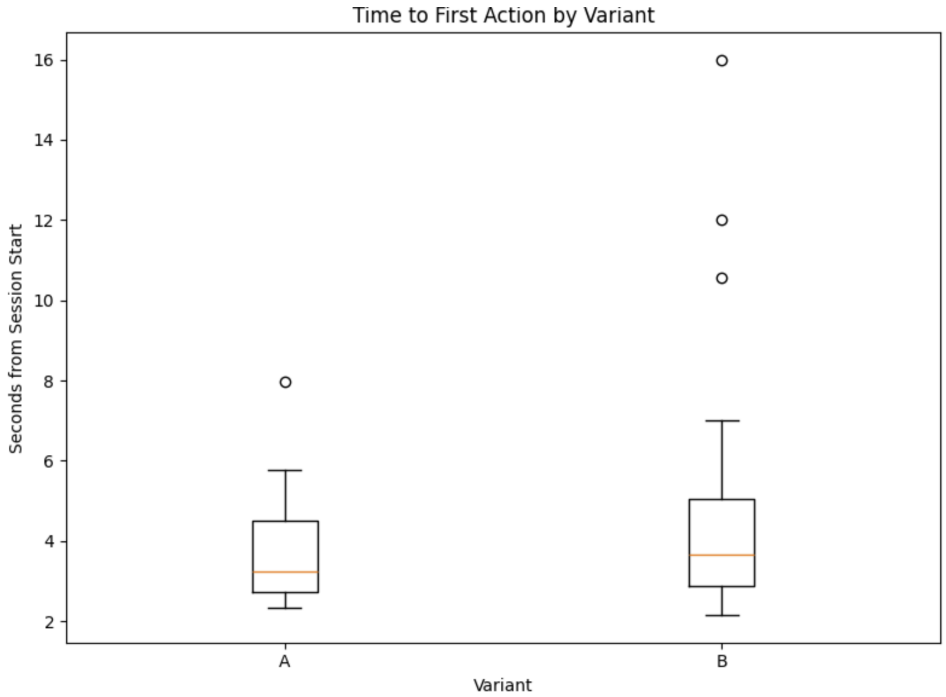

Figure 7: Time to First Action by Variant

Figure 7 compares the distribution of time to first action between the two variants. The median

time to first action appears relatively similar between the groups, although Version B shows several larger outliers.

Interpretation: Version B increased the probability that users completed a meaningful action, but it did not necessarily make all users act faster. Some Version B users took longer before completing their first action, which may explain the wider spread in the distribution.

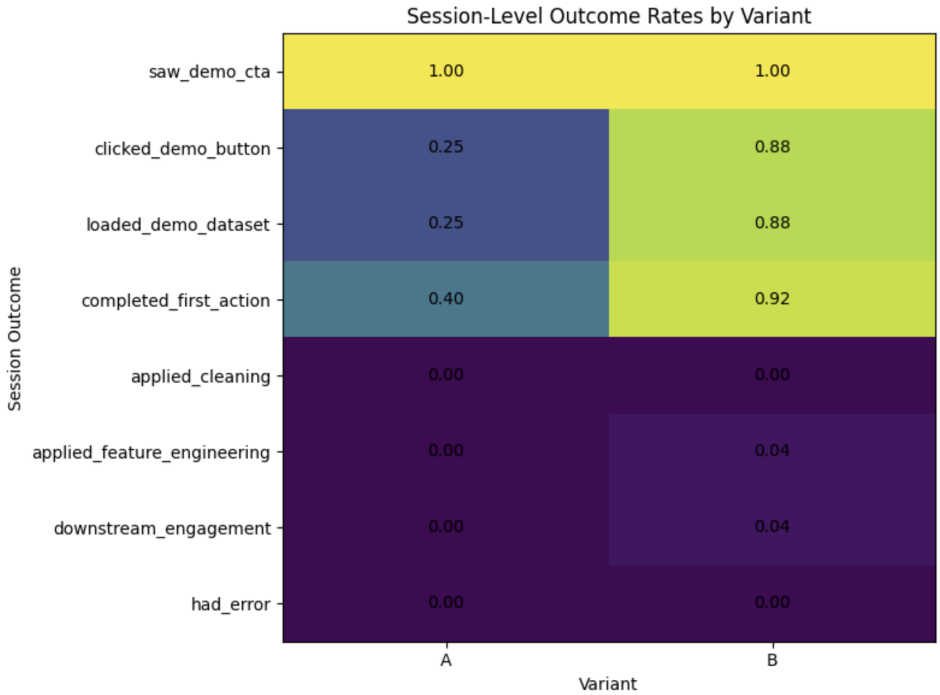

Figure 8: Session-Level Outcome Rates by Variant

Figure 8 summarizes session-level outcome rates across the two variants. Both groups had a demo CTA exposure rate of 1.00 because the analysis focused on CTA-exposed sessions. Version B had much higher rates for `clicked_demo_button`, `loaded_demo_dataset`, and `completed_first_action`.

Interpretation: The heatmap shows that Version B mainly improved early engagement outcomes, while deeper downstream actions such as cleaning, feature engineering, and downstream engagement remained low. This suggests that the treatment was effective at encouraging initial interaction but did not strongly affect later-stage app exploration.

5.2 Session Counts by Variant

The final session-level dataset included 44 CTA-exposed sessions. Among these sessions, 20 were assigned to Version A and 24 were assigned to Version B.

Table 3: Session Counts by Variant

Variant	Button Text	Number of Sessions
A	Load Demo Dataset	20
B	Try Demo Dataset Instantly	24

The two groups were reasonably comparable in size, although Version B had slightly more sessions than Version A. Because the sample size was relatively small, Fisher's exact test was used for the main binary outcome comparisons.

5.3 Raw Event Counts by Variant

Before constructing session-level outcomes, we examined the raw event counts by variant. These raw counts describe the total number of times each event occurred. They are useful for describing the overall interaction pattern, but they are not used directly as conversion rates because a single user session may trigger the same event more than once.

Table 4: Raw Event Counts by Variant

Event	Version A	Version B
`session_started`	20	24
`demo_cta_impression`	20	24
`demo_button_clicked`	6	27
`demo_dataset_loaded`	6	24
`first_action_completed`	8	22
`file_uploaded`	3	3
`eda_plot_generated`	0	21
`feature_engineering_applied`	0	1

The raw event counts show that Version B generated more interaction events than Version A, especially for demo button clicks, demo dataset loading, and EDA plot generation. However, because raw event counts may include repeated actions within the same session, the formal statistical analysis uses session-level indicators instead.

5.4 Session-Level Outcomes

For the formal analysis, the raw event data were aggregated to the session level. Each session was counted at most once for each outcome event. This prevents repeated clicks or repeated interactions within the same session from inflating the estimated engagement rates.

Table 5: Session-Level A/B Testing Summary by Variant

Variant	Sessions	Clicks	Loaded	First Actions	Downstream
A: Load Demo Dataset	20	5	5	8	0
B: Try Demo Dataset Instantly	24	21	21	22	1

Version B had more session-level demo clicks, demo dataset loads, and first action completions than Version A. No errors were recorded in either group.

6 Statistical Analysis and Results

6.1 Primary Metric

The primary metric was the **demo dataset load rate**, defined as the proportion of CTA-exposed sessions in which the user successfully loaded a demo dataset:

$$\text{Demo Dataset Load Rate} = \frac{\text{Number of sessions with } \texttt{demo_dataset_loaded}}{\text{Total number of CTA-exposed sessions}}.$$

This metric was selected because it directly measures whether the demo CTA encouraged users to begin using the app by loading a dataset.

The hypotheses were:

$$H_0 : p_A = p_B$$

$$H_1 : p_B > p_A$$

where p_A is the demo dataset load rate for Version A and p_B is the demo dataset load rate for Version B.

Table 6: Primary Metric: Demo Dataset Load Rate

Variant	**Successes**	**Total Sessions**	**Rate**
A: Load Demo Dataset	5	20	25.0%
B: Try Demo Dataset Instantly	21	24	87.5%

Version A had a demo dataset load rate of 25.0%, while Version B had a substantially higher rate of 87.5%. Fisher's exact test was used because the sample size was relatively small. The p-value was 3.18×10^{-5}, which is less than 0.05. Therefore, we reject the null hypothesis and conclude that Version B significantly improved the demo dataset load rate.

6.2 Secondary Metrics

The secondary metrics included click-through rate, first action completion rate, and downstream engagement rate. These metrics provide additional information about how the button text affected user behavior beyond the primary conversion outcome.

Table 7: Secondary Engagement Metrics by Variant

Metric	**Version A**	**Version B**	**p-value**
Click-through rate	25.0%	87.5%	3.18×10^{-5}
First action completion rate	40.0%	91.7%	0.000319
Downstream engagement rate	0.0%	4.17%	0.545

The click-through rate was 25.0% for Version A and 87.5% for Version B. This result was statistically significant using Fisher's exact test. In this dataset, the click-through rate and demo dataset load rate were the same at the session level, meaning that sessions with demo button clicks generally also triggered successful demo dataset loading.

The first action completion rate was also higher under Version B. Version A had a first action completion rate of 40.0%, while Version B had a rate of 91.7%. Fisher's exact test gave a p-value of 0.000319, indicating statistically significant evidence that Version B increased the probability of completing at least one meaningful action.

However, downstream engagement remained low in both groups. Version A had a downstream engagement rate of 0.0%, while Version B had a rate of 4.17%. The p-value was 0.545, so we do not reject the null hypothesis for this metric. This suggests that the treatment improved early-stage engagement, but there is not enough evidence that it improved deeper downstream interactions.

6.3 Statistical Testing Method

The formal statistical analysis was conducted using session-level binary outcomes. Although the raw GA4 data contain repeated event records, the analysis first converted the event-level data into session-level indicators so that each session was counted at most once for each outcome.

The primary outcome was whether a session successfully loaded a demo dataset. The secondary outcomes included whether a session clicked the demo dataset button, completed a first meaningful action, and showed downstream engagement.

Because the sample size was relatively small, Fisher's exact test was used to compare the outcome rates between Version A and Version B. The primary hypothesis was:

$$H_0 : p_A = p_B$$

$$H_1 : p_B > p_A$$

where p_A is the outcome rate for Version A and p_B is the outcome rate for Version B. The one-sided alternative was used because the treatment version was designed to increase user engagement.

6.4 Results Summary

The results are summarized in Table 8. Version B showed higher engagement than Version A on the primary metric and most secondary metrics.

Metric	**Version A**	**Version B**	**Difference**	**p-value**
Number of sessions	20	24	4	N/A
Demo dataset load rate	25.0%	87.5%	62.5 pp	3.18×10^{-5}
Click-through rate	25.0%	87.5%	62.5 pp	3.18×10^{-5}
First action completion rate	40.0%	91.7%	51.7 pp	0.000319
Downstream engagement rate	0.0%	4.17%	4.17 pp	0.545

Table 8: Summary of A/B testing results using session-level outcomes.

The primary metric, demo dataset load rate, increased from 25.0% in Version A to 87.5% in Version B. Fisher's exact test produced a p-value of 3.18×10^{-5}, so we reject the null hypothesis at the 5% significance level. This provides strong statistical evidence that Version B improved the demo dataset load rate.

For the secondary metrics, Version B also produced a higher click-through rate and first action completion rate. The click-through rate increased from 25.0% to 87.5%, and the first action completion rate increased from 40.0% to 91.7%. However, downstream engagement remained low in both groups, and the difference was not statistically significant ($p = 0.545$).

6.5 Interpretation

The A/B test provides strong evidence that the treatment CTA, "Try Demo Dataset Instantly," improved early user engagement. For the primary metric, Version B had a demo dataset load rate of 87.5%, compared with 25.0% for Version A. Fisher's exact test gave a p-value of 3.18×10^{-5}, which is far below the 0.05 significance level. Therefore, we reject the null hypothesis and conclude that Version B significantly improved the demo dataset load rate.

The secondary metrics show a similar pattern for early engagement. The click-through rate increased from 25.0% in Version A to 87.5% in Version B, and the first action completion rate increased from 40.0% to 91.7%. These results suggest that the more action-oriented button text encouraged users not only to click the demo dataset button, but also to complete at least one meaningful action in the app.

However, the treatment effect did not clearly extend to downstream engagement. The downstream engagement rate was 0.0% for Version A and 4.17% for Version B, with a p-value of 0.545. Since this p-value is greater than 0.05, we fail to reject the null hypothesis for downstream engagement. This means there is not enough statistical evidence to conclude that Version B increased deeper follow-up actions such as data cleaning, feature engineering, or other later-stage app interactions.

Overall, Version B was most effective at improving early-stage user engagement. It significantly increased demo dataset loading, click-through behavior, and first action completion, but it did not provide significant evidence of improving deeper downstream engagement.

7 Key Findings

The main findings from the A/B testing analysis are as follows:

1. **Google Analytics 4 successfully collected user interaction data from the Shiny application.**
   The GA4 event tracking setup allowed us to record user activity in the Data Explorer app, including sessions, button clicks, and other interaction events.

2. **The experiment successfully compared two versions of the demo dataset CTA button.**
   Version A used the original button text, "Load Demo Dataset," while Version B used the treatment text, "Try Demo Dataset Instantly." This setup allowed us to isolate the effect of the CTA wording change while keeping the rest of the app behavior the same across groups.

3. **Session-level aggregation was necessary for valid comparison.**
   Because raw GA4 data contains repeated event records from the same session, we aggregated the data to the session level before comparing Version A and Version B. This prevents repeated events from being treated as independent users.

4. **The primary metric focused on demo dataset loading.**
   Since the treatment changed the wording of the demo dataset button, the main outcome metric was the session-level demo dataset load rate. This metric directly measures whether users were more likely to begin using the app by loading a demo dataset after seeing the treatment wording. In this dataset, the click-through rate and demo dataset load rate were identical at the session level.

5. **Secondary metrics provided broader engagement evidence.**
   In addition to the primary CTA click metric, we compared secondary metrics such as total events, average events per session, average clicks per session, and downstream user actions. These metrics help evaluate whether the wording change affected not only the first click but also later interaction with the app.

6. **The results should be interpreted as exploratory.**
   Although the A/B testing framework was implemented successfully, the sample size and possible developer testing sessions limit the strength of the conclusion. Therefore, the findings should be interpreted as preliminary evidence rather than a fully conclusive product decision.

8 Limitations and Future Work

This project has several limitations. First, the sample size was relatively limited, so the estimated difference between Version A and Version B may not be fully stable. A/B testing usually requires a larger number of independent sessions to detect reliable differences. Second, some sessions may have come from developers testing the app during implementation, which may not fully represent real user behavior and could inflate event or click counts. Third, although we aggregated the raw Google Analytics event data to the session level, GA4 data can still be affected by repeated actions within the same session, browser differences, device type, and user familiarity with the app.

Another limitation is that this experiment only changed one interface element: the demo dataset CTA button text. This design helps isolate the effect of the wording change, but it also means the experiment does not evaluate other potentially important interface factors, such as button color, position, size, page instructions, or overall layout. Future work should collect more real user sessions over a longer period, apply stronger filtering to remove developer testing sessions, and test additional interface variations. It would also be useful to track more downstream actions, such as whether users continue to data preview, cleaning, feature engineering, or EDA after clicking the demo dataset button. These improvements would make the A/B testing results more reliable and provide stronger evidence for future app design decisions.

9 Contributions

- **Baixuan Chen (bc3212) — Data Cleaning, EDA, and A/B Test Analysis**
  Processed the exported Google Analytics 4 data, conducted exploratory data analysis, created visualizations, and performed the statistical comparison between Version A and Version B.

- **Lucas Meadows (lm3774) — Shiny App Development and Interface Design**
  Implemented and refined the main Shiny app structure, including the user interface components for data loading, demo dataset selection, and app navigation.

- **Shengbo Yi (sy3302) — Google Analytics Integration and Event Tracking**
  Added Google Analytics 4 tracking code and custom event tracking to collect user interaction data, including demo CTA clicks and other app events.

- **Wenyang Zu (wz2744) — Report Writing and GitHub Organization**
  Organized the GitHub repository, prepared project files for submission, helped maintain the README/report structure, and summarized project findings.

10 Conclusion

This project implemented an A/B testing framework in a Shiny-based Data Explorer application and used Google Analytics 4 to collect user interaction data. The experiment focused on one specific interface change: revising the demo dataset CTA button text from "Load Demo Dataset" in Version A to "Try Demo Dataset Instantly" in Version B.

The analysis compared user behavior between the two versions using session-level metrics. The primary metric focused on demo CTA engagement, while secondary metrics such as total events, average events per session, average clicks per session, and downstream user actions were used to evaluate broader engagement patterns. Aggregating the raw GA4 event data to the session level helped avoid treating repeated events from the same session as independent observations.

Overall, this project shows how web analytics and statistical testing can be combined to evaluate app design decisions. Since the sample size may be limited and some sessions may come from development testing, the results should be interpreted as exploratory rather than fully conclusive. With more real user sessions and a longer experiment period, the same A/B testing framework could provide stronger evidence for improving the Shiny app interface.

Appendix: Supporting Monitoring File

The file `datadog_consolidated_wide.csv` provides a supplementary wide-format summary of selected interaction events. It was used only as a supporting check of the tracking setup. The formal statistical analysis in this report was based on the GA4-derived session-level dataset.